Power of a Point

Problem 1. Let $ABCD$ be a cyclic quadrilateral with AB and CD not parallel. Let M be the midpoint of CD. Let P be a point inside $ABCD$ such that $PA = PB = CM$. Prove that AB, CD and the perpendicular bisector of MP are concurrent.

Problem 2. Let circles Ω and ω touch internally at point A. A chord BC of Ω touches ω at point K. Let O be the centre of ω. Prove that the circle BOC bisects segment AK.

Problem 3. Let ABC be an acute-angled triangle such that $AB < AC$. Let X and Y be points on the minor arc BC of the circumcircle of ABC such that $BX = XY = YC$. Suppose that there exists a point N on the segment AY such that $AB = AN = NC$. Prove that the line NC passes through the midpoint of the segment AX.

Problem 4. Points K and L are in the interior of a triangle ABC. Point D lies on the side AB. It is known that points B, K, L, C are concyclic, and $\angle AKD = \angle BCK$, $\angle ALD = \angle BCL$. Prove that $AK = AL$.

Problem 5. Let G be the centroid of a right-angled triangle ABC with $\angle BCA = 90°$. Let P be the point on ray AG such that $\angle CPA = \angle CAB$, and let Q be the point on ray BG such that $\angle CQB = \angle ABC$. Prove that the circumcircles of triangles AQG and BPG meet at a point on side AB.

Problem 6. In $\triangle ABC$, M is the midpoint of side AC. D, E are two points on the tangent line at A to the circumcircle of $\triangle ABC$, such that $MD // AB$, and A is the midpoint of DE. The circle passing through points A, B, E intersects the side AC again at P. The circle passing through points A, D, P intersects the extension of line DM again at Q. Prove that $\angle BCQ = \angle BAC$.

Problem 7. Let Γ be the circumcircle of triangle ABC. A circle Ω is tangent to the line segment AB and is tangent to Γ at a point lying on the same side of the line AB as C. The angle bisector of $\angle BCA$ intersects Ω at two different points P and Q. Prove that $\angle ABP = \angle QBC$.

Problem 8. Let ABC be a triangle, I its incentre and ω its incircle. Let D, E and F be the points of tangency of ω with BC, AC and AB, respectively and M, N and P be the midpoints of BC, AC and AB. Let D' be the second intersection of DI with ω, Q the intersection of DI with EF and $U \neq Q$ be the intersection of $(AD'Q)$ with (DMQ). Suppose that U lies on the circumcircle of BDF. Prove that PN, AM, UF concur.